Dataset Documentation — WTO Dispute Analysis

Last Updated: March 2026
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1. Panel Data

1.1 Country-Year Panel: `country_meta_1995_2024.csv`

Attribute	Value
Unit of analysis	Country-year
Rows	5,880
Countries / Polities	196
Year range	1995–2024 (30 years)
Columns	77
Primary key	`(country, year)`

The dataset integrates seven major data sources into a balanced country-year panel covering all WTO-relevant polities (WTO members, observers, and non-members that appear in WTO dispute records), plus the EU as a collective actor.

A. Identifiers

Variable	Definition	Unit / Range	Source
`country`	Country / polity name	String	User-defined
`iso3c`	ISO 3166-1 alpha-3 country code	3-letter string (e.g., `USA`, `TAW`)	WDI / User-defined
`ccode`	Correlates of War numeric country code	Integer; –1 if no COW match	COW State System
`stateabb`	COW state abbreviation (3-letter)	String (e.g., `USA`, `CHN`)	COW State System
`un_country_name`	Official UN country name	String	UN / WDI
`year`	Calendar year	Integer; 1995–2024	—
`COWcode`	COW numeric code (for V-Dem/NMC join)	Integer	COW / V-Dem

> ***Note on `ccode`:*** *Uses –1 as a sentinel for polities without a COW code (e.g., EU, Hong Kong, Macao, small island states).*

B. WTO Membership & Participation

Source: WTO official membership records and `wto_cases_v2.csv` dispute data (644 cases DS1–DS644, 1995–2024).

Variable	Definition	Unit / Range
`wto_member`	Whether the country is a WTO member in that year	Binary (0/1)
`wto_accession_year`	Year of WTO accession	Integer; 1995–2024; `NaN` for non-members (15.3%)
`complainant`	Country initiated at least one WTO dispute in that year	Binary (0/1)
`respondent`	Country was respondent in at least one dispute in that year	Binary (0/1)
`third_party`	Country joined as third party in at least one dispute in that year	Binary (0/1)
`wto_participant`	Any WTO dispute activity (complainant OR respondent OR third party)	Binary (0/1)
`cum_complainant`	Cumulative number of disputes initiated (1995–that year)	Integer; 0–25
`cum_respondent`	Cumulative number of disputes faced as respondent	Integer; 0–29
`cum_third_party`	Cumulative number of third-party appearances	Integer; 0–30

C. EU / Eurozone Membership

Source: European Union official membership records; European Central Bank Eurozone records.

Variable	Definition	Unit / Range
`eu_member`	Whether the country was an EU member in that year	Binary (0/1)
`euro_join_year`	Year the country adopted the Euro	Integer; 1999–2023; `NaN` for non-Euro countries (89.8%)
`euro_member`	Whether the country used the Euro in that year	Binary (0/1)

D. UN Voting Ideal Points

Source: Bailey, Michael A., Anton Strezhnev, and Erik Voeten. 2017. "Estimating Dynamic State Preferences from United Nations Voting Data." *Journal of Conflict Resolution* 61(2): 430–456.
Data version: July 28, 2025 (Erik Voeten, Georgetown University).

The ideal points are estimated using a Bayesian Item Response Theory (IRT) model fit to UNGA voting records. Higher values indicate alignment with the US; lower values indicate alignment with Russia/China.

Variable	Definition	Unit / Range
`idealpointfp`	Ideal point based on final-passage votes only	Continuous; approx. –2.1 to +3.0
`idealpointall`	Ideal point based on all votes	Continuous; approx. –2.4 to +3.1
`idealpointlegacy`	Ideal point from legacy session-based estimation	Continuous; approx. –2.2 to +3.2
`nvotesfp`	Number of final-passage votes	Integer; 1–109
`nvotesall`	Number of all votes	Integer; 1–232
`nvotesLegacy`	Number of votes for legacy estimation	Integer; 1–96
`qofp`	Posterior mean (FP)	Continuous
`q5fp`, `q10fp`, `q50fp`, `q90fp`, `q95fp`, `q100fp`	Posterior quantiles (FP)	Continuous
`qoall`	Posterior mean (all votes)	Continuous
`q5all`, `q10all`, `q50all`, `q90all`, `q95all`, `q100all`	Posterior quantiles (all votes)	Continuous

> ***ERGM datasets:*** *All `nvotes*`, `q*` posterior uncertainty columns are* ***excluded*** *from ERGM node attribute merges. Only the point estimates `idealpointfp`, `idealpointall`, `idealpointlegacy` are retained (suffixed `_1` / `_2`). Taiwan/HK/Macao/EU have NaN for all ideal point columns.*

E. V-Dem Democracy Indicators

Source: Coppedge, Michael, et al. 2025. "V-Dem Codebook v15." Varieties of Democracy (V-Dem) Project, University of Gothenburg.

Variable	Definition	Unit / Range
`v2x_polyarchy`	Electoral Democracy Index	Continuous; 0–1

Variable	Definition	Unit / Range
`v2x_libdem`	Liberal Democracy Index	Continuous; 0–1
`v2x_regime`	Regime type classification	0=Closed Autocracy; 1=Electoral Autocracy; 2=Electoral Democracy; 3=Liberal Democracy
`v2eltype_0`	Legislative election held (first round)	0/1; `NaN` in non-election years (74.7%)
`v2eltype_1`	Legislative election held (second round)	0/1; `NaN` in non-election years
`v2eltype_6`	Presidential election held (first round)	0/1; `NaN` in non-election years
`v2eltype_7`	Presidential election held (second round)	0/1; `NaN` in non-election years
`is_leg_elec`	Any legislative election in that year	Binary (0/1); derived
`is_pres_elec`	Any presidential election in that year	Binary (0/1); derived
`election_binary`	Any election (legislative or presidential)	Binary (0/1); derived

> ***Note:*** *`v2eltype_*` are `NaN` in non-election years — this is by design, not missing data.*
>
> ***ERGM datasets:*** *`v2eltype_0`, `v2eltype_1`, `v2eltype_6`, `v2eltype_7`, `is_leg_elec`, `is_pres_elec` are **excluded** from ERGM node attribute merges. Only `v2x_polyarchy`, `v2x_libdem`, `v2x_regime`, and `election_binary` are retained.*

F. World Development Indicators (WDI)

Source: World Bank, World Development Indicators (WDI). Accessed 2025.

Unit standardization: Monetary values in **Million USD**. Percentage values divided by 100 to **Ratio (0–1)**.

Variable	WDI Code	Definition	Unit
`gdp`	`NY.GDP.MKTP.KD`	Real GDP (constant 2015 USD)	Million USD
`gdppc`	`NY.GDP.PCAP.KD`	Real GDP per capita (constant 2015 USD)	USD

`pop`	`SP.POP.TOTL`	Total population	Count
`gdp_growth_rate`	`NY.GDP.MKTP.KD.ZG`	Annual GDP growth rate	Ratio
`trade`	`NE.TRD.GNFS.ZS`	Total trade (exports + imports) as % of GDP	Ratio
`fdi`	`BX.KLT.DINV.WD.GD.ZS`	Net FDI inflows as % of GDP	Ratio

`fdi_inflow_usd`	`BX.KLT.DINV.CD.WD`	Net FDI inflows (absolute)	Million USD
`exp_share`	`NE.EXP.GNFS.ZS`	Exports of goods and services as % of GDP	Ratio
`imp_share`	`NE.IMP.GNFS.ZS`	Imports of goods and services as % of GDP	Ratio
`unemployment_rate`	`SL.UEM.TOTL.NE.ZS`	Unemployment rate (national estimate)	Ratio

> ***ERGM datasets:*** *`gdp` (total) is **excluded** from ERGM node attribute merges. Use `gdppc` and `log_gdppc` instead. All other WDI columns above are retained.*

G. Worldwide Governance Indicators (WGI)

Source: Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi. 2010. "The Worldwide Governance Indicators: Methodology and Analytical Issues." *World Bank Policy Research Working Paper* No. 5430.

All six WGI indicators are on a standard normal scale (−2.5 to +2.5), higher = better governance.

Variable	WGI Code	Definition
`voice`	`VA.EST`	Voice and Accountability
`stability`	`PV.EST`	Political Stability and Absence of Violence/Terrorism
`efficiency`	`GE.EST`	Government Effectiveness
`reg_quality`	`RQ.EST`	Regulatory Quality
`law`	`RL.EST`	Rule of Law
`corruption`	`CC.EST`	Control of Corruption

WGI interpolation: 1995 backfilled from 1996; 1997, 1999, 2001 linearly interpolated from adjacent survey years.

H. Geographic & Classification Variables

Source: World Bank WDI.

Variable	Definition	Values
`region`	World Bank geographic region	7 regions; `NaN` for EU
`longitude`	Geographic centroid longitude	Decimal degrees
`latitude`	Geographic centroid latitude	Decimal degrees
`income`	World Bank income group	`Low income`, `Lower middle income`, `Upper middle income`, `High income`

I. Derived Variables

Variable	Definition	Formula
`log_gdppc`	Natural log of real GDP per capita	`ln(gdppc)`
`log_pop`	Natural log of total population	`ln(pop)`

J. COW National Material Capabilities (NMC)

Source: Singer, Bremer, and Stuckey. 1972. **Dataset version:** NMC 6.0 (1816–2016).

> ***Coverage note:*** *NMC v6.0 covers* ***1816–2016 only****. All observations for* ***2017–2024 are*** `NaN` *(1,568 missing observations).*

Variable	Definition	Unit
`cinc`	Composite Index of National Capability	Proportion; 0–1
`milex`	Military expenditures	Thousands USD
`milper`	Military personnel	Thousands of persons
`pec`	Primary energy consumption	Thousands of coal-ton equivalents
`tpop`	Total population	Thousands of persons
`upop`	Urban population (cities >100k)	Thousands of persons

> ***Sentinel values:*** *COW uses* ***−9*** *to denote missing values. These are retained as-is; recode to* `NaN` *before modeling.*
>
> ***ERGM datasets:*** *Only* `cinc` *is retained from NMC.* `milex`*,* `milper`*,* `pec`*,* `tpop`*,* `upop` *are* ***excluded****.* `cinc` *is NaN for all years 2017–2024 (NMC v6.0 ends 2016).*

1.2 WTO Participation Panel: `wto_panel_1995_2024.csv`

Attribute	Value
Unit of analysis	Country-year
Rows	5,880
Columns	18

The WTO-specific subset of `country_meta_1995_2024.csv`. Contains only identifiers (`country`, `iso3c`, `ccode`, `stateabb`, `un_country_name`, `year`), WTO participation variables (`wto_member`, `wto_accession_year`, `complainant`, `respondent`, `third_party`, `wto_participant`,

`cum_complainant`, `cum_respondent`, `cum_third_party`), and EU/Eurozone membership (`eu_member`, `euro_join_year`, `euro_member`).

1.3 WTO Membership List: `wto_mem_list.csv`

Attribute	Value
Rows	166
Columns	5

Variable	Definition
`member`	Member name (WTO official)
`membership_date`	Full accession date
`iso3c`	ISO3 code
`ccode`	COW numeric code
`year`	Accession year (integer)

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2. Dyadic Data

2.1 WTO Dispute Dyads: `wto_dyadic_v2.csv`

Attribute	Value
Unit of analysis	Case-level directed dyad
Rows	8,373
Columns	36
Unique cases	644

Each row represents a directed country-pair relationship within a specific WTO dispute case. Country names harmonized: European Communities → European Union; Turkey → Türkiye.

Variable	Definition
`case`	WTO case number (DS1–DS644, string format "DS{N}")
`country1`	First country in the dyad
`country2`	Second country in the dyad
`relationship`	

Variable	Definition
	`complainant-respondent` (684), `complainant-third_party` (3,946), or `third_party-respondent` (3,743)
`iso3_1`, `iso3_2`	ISO3 codes for each country
`ccode1`, `ccode2`	COW codes for each country
Remaining 28 columns	Full case metadata inherited from `wto_cases_v2.csv` (title, short_title, summary, all procedural dates, `dispute_stage`, `product`)

EU handling note: 395 cases have EUN-only; 37 cases list both EUN and individual member states (e.g., DS316 lists USA vs EUN, FRA, DEU, ESP, GBR separately); 30 early cases list individual EU member states without EUN (e.g., DS37: USA vs PRT). See Section 5.7 for EU disaggregation rules.

2.2 Bilateral Trade (WTO-filtered): `bilateral_trade_wto.csv`

Attribute	Value
Unit of analysis	Directed dyad-year (exporter → importer)
Rows	545,333
Columns	31
Years	1995–2024
Scripts	`build_baci_trade.py` (base trade) + `build_dyadic_datasets.py` (merge & filter)

Directed dyad-year bilateral trade data filtered to pairs where **both exporter and importer are WTO members** in the given year (time-varying membership filter). Merges base BACI trade data (882,643 rows) with ATOP alliance data and DESTA trade agreement data, then applies WTO membership filtering (removes 337,310 non-WTO-member dyad-year observations).

Uses **dual EU representation**: individual EU member states retain their own trade flows (including intra-EU trade) while a separate `EUN` aggregate captures the EU as a single external trade actor (intra-EU trade excluded).

Trade Variables (from BACI HS92)

Variable	Definition	Unit
`year`	Calendar year	Integer; 1995–2024
`exporter`	Exporter ISO3 code	String
`exporter_ccode`	Exporter COW numeric code	Integer

Variable	Definition	Unit
`exporter_name`	Exporter country name	String
`importer`	Importer ISO3 code	String
`importer_ccode`	Importer COW numeric code	Integer
`importer_name`	Importer country name	String
`total_trade_ij`	Total exports from i to j	Thousands USD
`n_products_ij`	Number of distinct HS6 products traded	Integer
`n_sections_ij`	Number of distinct HS sections traded	Integer; 1–21
`export_dependence`	`total_trade_ij / total_exports_i` — Partner dependence	Ratio (0–1)
`trade_dependence`	`(exports_ij + imports_ij) / gdp_i` — Bilateral trade relative to GDP	Ratio; unbounded
`total_exports_i`	Total exports of exporter i (all destinations)	Thousands USD

> ***Import data omitted by design:*** *Import-side columns are not stored because they are redundant — the import flow from j to i equals the export flow in the reverse row (j, i). Reconstruct if needed by joining on the reverse direction.*
>
> ***`trade_dependence`:*** *Can exceed 1 for small open economies (e.g., ship-registration economies).* `NaN` *for entities without GDP in the panel (2.4%).*

ATOP Alliance Variables

Source: Leeds et al. 2002. ATOP v5.1 (1815–2018). Forward-filled 2019–2024 from 2018.

Variable	Definition	Values
`atopally`	Any alliance between exporter and importer	0/1
`defense`	Exporter has defensive obligation toward importer	0/1
`offense`	Exporter has offensive obligation toward importer	0/1
`neutral`	Neutrality obligation	0/1
`nonagg`	Non-aggression pact	0/1
`consul`	Consultation obligation	0/1

> ***ATOP uses COW ccodes.*** *After WTO membership filtering, all rows have valid ccodes on both sides, so no ATOP* `NaN` *values remain. Allied: 130,206 (23.9%). Not allied: 415,127 (76.1%).*

DESTA Trade Agreement Variables

Source: Dur, Baccini, and Elsig. 2014. DESTA v2.

Variable	Definition	Values
`label`	Any trade agreement in force between dyad	0/1
`number`	DESTA agreement identifier	Integer; `NaN` if no agreement
`base_treaty`	Base treaty number	Integer
`name`	Agreement name	String
`entry_type`	Type of entry into the agreement	Categorical
`typememb`	Membership type (1=bilateral, 2=plurilateral, etc.)	Integer; 1–7
`depth_index`	Additive depth index (7 provisions)	Integer; 0–7
`depth_rasch`	Rasch-scaled depth measure (latent trait)	Continuous
`flexrigid`	Flexibility/rigidity provisions	Continuous
`flexescape`	Escape clause provisions	Continuous
`enforce`	Dispute settlement mechanism strength (raw)	Continuous
`enforce01`	Standardized DSM strength index	Continuous; 0–6

> ***DESTA is undirected.*** *The pipeline merges both directions to ensure CHN→AUS gets the same values as AUS→CHN. Has agreement: 205,130 (37.6%). No agreement: 340,203 (62.4%).*

2.3 Bilateral Trade by Section (WTO-filtered): `bilateral_trade_section_wto.csv`

Attribute	Value
Unit of analysis	Directed dyad-year-section
Rows	6,374,486
Columns	14
Sections	21 HS sections
Scripts	`build_baci_trade.py` (base) + `build_dyadic_datasets.py` (WTO filter)

Section-level breakdown of bilateral trade, WTO-filtered. No ATOP/DESTA columns at this level.

Variable	Definition	Unit
`year`	Calendar year	

Variable	Definition	Unit
		Integer; 1995–2024
`exporter`	Exporter ISO3 code	String
`exporter_ccode`	Exporter COW numeric code	Integer
`importer`	Importer ISO3 code	String
`importer_ccode`	Importer COW numeric code	Integer
`section_num`	HS section number	Integer; 1–21
`section_en`	HS section English name	String
`trade_value_ij_s`	Exports from i to j in section s	Thousands USD
`n_products_ij_s`	Number of distinct HS6 products in section s	Integer
`bilateral_sector_concentration`	`trade_value_ij_s / total_exports_i` — How important is section s for i→j trade	Ratio (0–1)
`sector_export_concentration`	`total_exports_i_s / total_exports_i` — How important is section s for country i	Ratio (0–1)
`partner_sector_dependence`	`trade_value_ij_s / total_exports_i_s` — How much does i rely on j for section s	Ratio (0–1)
`total_exports_i`	Total exports of exporter i (all destinations, all sections)	Thousands USD
`total_exports_i_s`	Total exports of exporter i in section s (all destinations)	Thousands USD

> ***No missing values*** *in this dataset.*

2.4 DESTA Trade Agreement Panel: `desta_panel_1995_2025.csv`

Attribute	Value
Unit of analysis	Undirected dyad-year (country1 < country2 alphabetically)
Rows	235,011
Columns	21
Years	1995–2025
Unique dyads	7,581
Script	`build_dyadic_datasets.py`

Variable	Definition
`country1`, `country2`	Country names (alphabetically sorted)
`year`	Calendar year
`label`	Trade agreement in force (0/1). Has agreement: 167,521; No agreement: 67,490
`iso3_1`, `iso3_2`	ISO3 codes (mapped from ISO 3166-1 numeric via `country_converter`)
`baci_code1`, `baci_code2`	ISO 3166-1 numeric codes (from original DESTA data)
`ccode1`, `ccode2`	COW numeric codes
`number`, `base_treaty`, `name`	DESTA agreement identifiers
`entry_type`, `typememb`	Agreement membership type
`depth_index`, `depth_rasch`	Depth measures
`flexrigid`, `flexescape`	Flexibility provisions
`enforce`, `enforce01`	Dispute settlement strength

> ***ISO mapping note:*** *DESTA uses ISO 3166-1 numeric codes (e.g., USA=840, France=250) which differ from BACI codes (USA=842, France=251). The pipeline maps these using* `country_converter` *with* `src='ISOnumeric'`*. Taiwan (ISO=158) is mapped to* `TAW` *to match the panel dataset. Unmapped iso3_1: 651 rows (mostly dissolved states).*

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3. WTO Case Data

3.1 Case Dataset: `wto_cases_v2.csv`

Attribute	Value
Rows	644 (DS1–DS644)
Columns	29
Script	`scripts/scrape_wto_cases.py` (scraped March 2026)

Country names harmonized: European Communities → European Union (230 cases); Turkey → Türkiye (119 cases).

Variable	Definition
`case`	Case number (string "DS{N}")
`title`	Full case title ("Respondent — Product" format)
`short_title`	Short case name
`complainant`	Complainant country/entity name
`respondent`	Respondent country/entity name
`third_parties`	Semicolon-separated list of third-party countries
`product`	Product/subject extracted from title
`dispute_stage`	6 categories: Consultation, Mutually Agreed Solution, Panel, Appellate Body, Implementation, Retaliation
`current_status`	Free-text current status description
`summary`	Case summary text
`agreements_cited_consultations`	WTO agreements cited at consultation stage
`agreements_cited_panel`	WTO agreements cited at panel stage
`consultations_requested`	Date consultations requested
`panel_requested`	Date panel requested
`mutually_agreed_solution_notified`	Date of mutually agreed solution
`panel_established`	Date panel established
`panel_composed`	Date panel composed
`panel_report_circulated`	Date panel report circulated
`appellate_body_report_circulated`	Date appellate body report circulated
`Art 21.5 Panel requested`	Date Art 21.5 panel requested
`Art 21.5 DSU Panel report circulated`	Date Art 21.5 panel report circulated
`Art 21.5 DSU Appellate Body report circulated`	Date Art 21.5 AB report circulated
`Art 22.6 DSU Arbitration decision circulated`	Date Art 22.6 decision
`Art 25 DSU Arbitration award circulated`	Date Art 25 award
`Art 21.5 DSU Panel lapsed`	Date Art 21.5 panel lapsed
`panel_lapsed`	Date panel lapsed
`arbitration_agreement`	Arbitration agreement details

Variable	Definition
`recourse_to_arbitration`	Recourse to arbitration details
`article_25_arbitrator_composed`	Date Art 25 arbitrator composed

Dispute Stage Distribution:

Stage	Cases
Consultation	~239
Panel	~178
Appellate Body	~115
Mutually Agreed Solution	~69
Implementation	~29
Retaliation	~14

> *Note: DS627–DS644 (18 cases) have metadata but no PDF documents. Severity scoring and HS classification cover only DS1–DS626.*

3.2 RAG Pipeline Outputs (`Data/Output/`)

All files cover DS1–DS626 (626 cases with PDF documents). Cases DS627–DS644 have no RAG outputs.

`industry_extraction.csv`

Attribute	Value
Rows	626
Columns	11

Variable	Definition
`case_id`	Integer case number
`case_title`	Case title
`title_product`	Product extracted from title
`product_descriptions`	RAG-extracted product descriptions
`explicit_hs_codes`	Explicitly cited HS codes in documents
`is_systemic`	Whether dispute concerns a systemic policy measure

Variable	Definition
`is_services`	Whether dispute concerns services (GATS)
`confidence`	Extraction confidence
`notes`	Additional notes
`retrieved_context`	Retrieved document context (truncated)
`n_parents_retrieved`	Number of parent chunks retrieved

`case_hs_sections.csv`

Attribute	Value
Rows	626
Columns	12

Variable	Definition
`case_id`	Integer case number
`case_title`	Case title
`product`	Product from `wto_cases_v2.csv` (ground truth)
`title_hs_sections`	HS sections from product/title (pipe-separated integers 1–21)
`title_case_type`	Case type from title: `product` or `policy`
`hs_sections`	HS sections from RAG + LLM classification (pipe-separated integers 1–21)
`case_type`	LLM-assigned case type: `product` or `policy`
`product_descriptions`	Product descriptions (product cases only)
`policy`	Policy description (policy cases only)
`extraction_method`	`explicit_hs`, `llm_classification`, or `title_fallback`
`confidence`	Classification confidence
`reasoning`	LLM reasoning

`case_section_expanded.csv`

Attribute	Value
Rows	3,003
Columns	5
Primary key	`(case_id, hs_section)`

One row per case-section pair. Policy cases with horizontal measures span all 21 sections (`is_horizontal_policy = True`).

Variable	Definition
`case_id`	Integer case number
`hs_section`	HS section number (integer 1–21)
`case_type`	`product` or `policy`
`extraction_method`	Method used for this classification
`confidence`	Confidence level

`severity_scores_raw.csv`

Attribute	Value
Rows	626
Columns	10

Raw (un-normalized) severity scores. Normalized version deferred until third-party scoring is complete.

Variable	Definition	Range
`case_id`	Integer case number	1–626
`complainant`	Complainant country name	—
`rhetorical_aggressiveness`	1=procedural/hedged → 5=hostile/geopolitical	1–5
`systemic_reach`	1=product-specific → 5=regime-challenging	1–5
`escalation_ultimatum`	1=routine → 5=retaliatory/rebalancing	1–5
`domestic_victimhood`	1=no domestic pain → 5=existential threat	1–5
`severity_score`	Mean of 4 dimensions (composite)	1–5
`reasoning`	LLM reasoning for scores	—
`evidence`	Supporting evidence text	—
`n_parents_retrieved`	Parent chunks used for scoring	Integer

> *Scoring anchors: DS3 = 1 (procedural/hedged), DS267 = 3 (assertive/data-heavy), DS574 = 5 (geopolitical/aggressive)*

3.3 WTO Dispute Analysis Results

Network Analysis Results (`wto_analysis_results_1995-2024.json`)

Annual cumulative network metrics (conflict-weighted dispute graph):

Year	Nodes	Edges	Conflict Density	Modularity	Communities
1995	26	100	0.086	0.196	3
2000	36	267	0.065	0.019	3
2005	26	100	0.037	0.182	4
2010	33	217	0.036	0.031	2
2015	32	227	0.026	−0.052	3
2018	50	1,117	0.032	0.086	4
2020	23	87	0.020	0.173	2
2024	20	39	0.026	0.200	1

Document Processing Statistics (March 2026 — Final)

Metric	Value
Total PDFs processed	9,417 (63 pre-split PDFs manually combined)
Successfully classified	9,414 (99.97%)
Manual review required	3 files (non-English cross-references: `293-19.pdf`)
Document types	42 types
Date coverage	~90% (multilingual extraction + multi-part inheritance)
Third-party joinings detected	1,036 entries

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4. Systematic Missing Data

4.1 Panel Dataset (`country_meta_1995_2024.csv`)

Overall missing data by variable group:

Variable Group	Missing %	Reason
`wto_accession_year`	15.3%	Non-WTO members
`euro_join_year`	89.8%	Non-Euro countries

Variable Group	Missing %	Reason
`idealpointfp`	4.1%	Non-UN members/observers
`idealpointall` / `idealpointlegacy`	7.4%	Same + broader vote gaps
`COWcode` / V-Dem indices	12.6%	Entities outside COW/V-Dem
`v2eltype_*`	74.7%	`NaN` in non-election years (by design)
`gdp`, `gdppc`	4.4%	Non-reporting states
`trade`, `exp_share`, `imp_share`	18.1%	Small/conflict states
`unemployment_rate`	42.8%	Many developing countries don't report
`cinc` / NMC variables	28.7%	NMC v6.0 ends 2016 (all countries `NaN` 2017–2024)
WGI (`voice`, `stability`, etc.)	5.6–8.6%	Small/non-covered states

Countries systematically missing from each source:

Source	Missing Polities	Reason
UN Voting (all years)	Taiwan, Hong Kong, Macao, EU	Not UN members
UN Voting (partial)	Afghanistan, Bosnia, Burundi, Cambodia, etc. (30+ countries)	Non-membership or irregular participation in some years
V-Dem (all years)	EU, Hong Kong, Macao, Andorra, Monaco, San Marino, Liechtenstein, Brunei, 8 Pacific island states, 9 Caribbean island states, South Sudan, Montenegro	Supranational entities, micro-states, SARs
COW NMC (2017–2024)	All 196 countries	NMC v6.0 coverage ends 2016
COW NMC (1995–2016)	EU, Hong Kong, Macao, Andorra, Monaco, San Marino, Liechtenstein	Not sovereign COW states
WDI	North Korea, Syria (post-2011), Somalia, South Sudan (pre-2011), EU, Taiwan (supplemented)	Sanctions, conflict, non-reporting
WGI	North Korea, very small island states, EU	Not covered

4.2 Bilateral Trade Dataset (`bilateral_trade_wto.csv`)

Variable	Missing %	Reason
`trade_dependence`	2.4%	Entities without GDP in panel (e.g., EUN, territories)
`number` , `base_treaty` , `name` , `entry_type` , `typememb`	56.5%	Dyads with no DESTA trade agreement (label=0)
`depth_index` , `depth_rasch` , `flexrigid` , `flexescape` , `enforce` , `enforce01`	62.4%	No agreement (label=0) or agreement exists but no depth coding

> *All other variables (trade, ATOP) have 0% missing after WTO membership filtering.*
>
> ***EUN DESTA note:*** *All 3,754 EUN rows in this file have* `label=0` *. This is a data artifact — DESTA records EU trade agreements through individual member states (e.g., EU-Korea FTA appears as DEU→KOR). The ERGM pipeline fixes this post-hoc;* `bilateral_trade_wto.csv` *itself is not modified.*

4.3 Bilateral Trade Section Dataset (`bilateral_trade_section_wto.csv`)

No missing values in any column.

4.4 ERGM Datasets

Systematic missing data in `ergm_dyad_year_eun.csv` and `ergm_dyad_year_eu_disagg.csv` :

Variable Group	Affected Rows / %	Reason
Trade lags `*_t1/t2/t3`	1995 rows: 100% NaN; 1996: t-2/t-3 imputed; 1997: t-3 imputed	No pre-1995 BACI data; backward imputation applied (see Section 5.5)
UN ideal points `idealpointfp_*` , `idealpointall_*` , `idealpointlegacy_*`	Taiwan (TAW), HK (HKG), Macao (MAC), EU (EUN): all years NaN	Not UN members
V-Dem `v2x_polyarchy_*` , `v2x_libdem_*` , `v2x_regime_*`	EU, HK, MAC, micro-states: all years NaN	Outside V-Dem coverage
CINC `cinc_*`	All countries 2017–2024: NaN	NMC v6.0 ends 2016
WGI `voice_*` , `stability_*` , etc.	EUN before aggregation; North Korea; very small states	Not covered by WGI

Variable Group	Affected Rows / %	Reason
WDI `gdppc_*`, `log_gdppc_*`, `pop_*`	~4.4% NaN	Non-reporting states (North Korea, Syria, Somalia); EUN filled by member-state aggregation
Severity `severity_max`, `severity_avg`, `dyadic_severity_*`	DS627–DS644 (18 cases): NaN	No PDF documents; RAG pipeline covers DS1–DS626 only
TP-R severity (`dyadic_severity` = 0.5)	All `third_party-respondent` rows	Placeholder pending third-party scoring completion
Disputed sector trade `disputed_trade_*`	Rows without disputes: NaN (by design)	Only populated for has_dispute=1 rows

> ***ERGM behavior:*** *Most ERGM implementations (R `ergm`, `btergm`) drop nodes/edges with missing covariates. For cross-sectional ERGM at a single year, restrict to that year's rows and assess missing data before modeling.*

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5. Data Processing & Construction Notes

5.1 Unit Standardization

Transformation	Details
GDP / FDI monetary values	Divided by 1,000,000 → Million USD
WDI percentage variables	Divided by 100 → Ratio (0–1)
COW NMC sentinel values	Raw −9 values retained; recode to `NaN` before use
BACI trade values	Thousands of current USD (original BACI units)

5.2 WGI Interpolation

WGI surveys were not conducted annually before 2002:
- **1995**: Backfilled using 1996 data.
- **1997**: Linear interpolation of 1996 and 1998.
- **1999**: Linear interpolation of 1998 and 2000.
- **2001**: Linear interpolation of 2000 and 2002.

5.3 WTO Participation Variables

- `complainant`, `respondent`, `third_party` are binary indicators constructed from `wto_cases_v2.csv` (644 cases DS1–DS644): 1 if the country participated in at least one dispute of that type in that year, 0 otherwise.

- `cum_complainant`, `cum_respondent`, `cum_third_party` are cumulative counts from 1995 through year t (inclusive).
- The EU is treated as a single litigation actor (`EUN`). Individual EU member states may also appear separately in early cases (see Section 5.7).
- **ERGM note:** For the ERGM datasets, annual activity counts (`n_complainant_t`, `n_respondent_t`, `n_tp_t`) are computed directly from `wto_dyadic_v2.csv` and merged alongside the cumulative `cum_*` variables. The annual counts capture how many distinct cases a country was involved in that specific year.

5.4 V-Dem Election Variables

- `v2eltype_*` variables are `NaN` in non-election years (not missing data).
- `is_leg_elec`, `is_pres_elec`, `election_binary` are derived aggregates.

5.5 Bilateral Trade Pipeline

```
BACI HS92 raw data → build_baci_trade.py → bilateral_trade_aggregate.csv (882,643 rows)
                                          → bilateral_trade_by_section.csv (8.9M rows)
                                                ↓
                      build_dyadic_datasets.py:
                        + ATOP 5.1 alliances (forward-filled 2019-2024)
                        + DESTA trade agreements (both merge directions)
                        + WTO membership filter (time-varying)
                                                ↓
                      bilateral_trade_wto.csv (545,333 rows)
                      bilateral_trade_section_wto.csv (6,374,486 rows)
```

WTO membership filter: Removes rows where either exporter or importer was not yet a WTO member in the given year. Reduces aggregate from 882,643 to 545,333 rows (−38.2%) and section-level from 8,880,578 to 6,374,486 rows (−28.2%).

DESTA merge fix: DESTA is undirected (country1 < country2 alphabetically). The pipeline merges both directions (exporter=iso3_1/importer=iso3_2 AND exporter=iso3_2/importer=iso3_1) to ensure symmetric coverage.

DESTA ISO code fix: DESTA uses ISO 3166-1 numeric codes which differ from BACI codes for 15 countries (e.g., USA: ISO=840, BACI=842; France: ISO=250, BACI=251). Fixed using `country_converter` with `src='ISOnumeric'`.

5.6 EU Representation in Trade Data

Entity type	Rows include	`total_exports_i` basis	Intra-EU trade
Individual EU member (e.g., `FRA`)	All bilateral flows including intra-EU	All exports including to other EU members	Included

Entity type	Rows include	`total_exports_i` basis	Intra-EU trade
EU aggregate (`EUN`)	Only EU external flows	EU external exports only	Excluded

Belgium-Luxembourg: BACI reports as a single unit (code 58) until 1998, mapped to Belgium.

EUN DESTA data gap: DESTA records EU trade agreements through individual member state pairs (e.g., the EU-Korea FTA appears as DEU→KOR "EC Korea"). As a result, all EUN rows in `bilateral_trade_wto.csv` have `label=0`. The ERGM pipeline (`build_ergm_data.py`) fixes this by propagating `label=1` to EUN rows whenever any EU member state has a trade agreement with the partner country in that year. The fix is applied in `ergm_dyad_year_eun.csv` and `ergm_dyad_year_eu_disagg.csv` only; `bilateral_trade_wto.csv` itself is unchanged.

EUN ATOP data: EUN has `atopally=0` for all rows. This is correct — the European Union as a collective entity is not an ATOP signatory. Individual EU member states (e.g., DEU, FRA) have their own ATOP entries (primarily NATO membership).

EUN node attributes: EUN has NaN for all economic (WDI), political (V-Dem, UN Voting, WGI), and military (NMC) variables in `country_meta_1995_2024.csv`. Only WTO participation variables (`wto_member` , `cum_*`) are populated. The ERGM pipeline aggregates EUN attributes from individual EU member states: sum for `pop` ; GDP-weighted average for all other numeric variables.

5.7 EU Disaggregation Rules for ERGM Datasets

Three categories of EU involvement in `wto_dyadic_v2.csv` require different handling in `ergm_dyad_year_eu_disagg.csv` :

Category	Count	Example	Rule
EUN-only — only EUN is listed as party	395 cases	DS2 (USA vs EUN)	Expand EUN → all individual EU member states in that year from `eu_membership_1995_2024.csv` . Dispute flags propagated to each member's bilateral_trade_wto row.
Mixed — both EUN and individual EU members listed	37 cases	DS316 (Airbus: USA vs EUN, FRA, DEU, ESP, GBR)	Drop EUN rows; keep individual member rows. They are already present and correctly populated.
Individual-EU-only — individual EU members listed, no EUN	30 cases	DS37 (USA vs PRT)	Keep as-is. These reflect early WTO practice before EU was consistently a single actor.

> ***Interpretation note:*** *EU disaggregation is a robustness check. The primary analysis should use* `ergm_dyad_year_eun.csv` *(EUN as unitary actor), which reflects how WTO disputes are actually filed.*

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6. Reference Files

6.1 BACI Code Mapping: `baci_to_iso3_mapping.csv`

Variable	Definition
`baci_code`	BACI numeric country code
`iso3`	ISO alpha-3 country code
`ccode`	COW numeric code
`country_name`	Country name from BACI metadata

6.2 EU Membership Panel: `eu_membership_1995_2024.csv`

Attribute	Value
Rows	840 (28 EU members x 30 years)
Columns	8

Variable	Definition
`country`	Country name
`iso3c`	ISO alpha-3 code
`ccode`	COW numeric code
`baci_code`	BACI numeric code
`year`	Calendar year
`eu_member`	Whether the country was an EU member in that year (0/1)
`accession_year`	Year of EU accession
`exit_year`	Year of EU exit; `NaN` if still member

6.3 Harmonized Case Dataset: `wto_cases_harmonized.csv`

Harmonized version of case data with standardized country names for network analysis.

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7. ERGM Analysis Datasets

Constructed by `scripts/build_ergm_data.py` and `DataConstruction.ipynb`. See README for full methodology.

7.1 `wto_cases_enriched.csv`

Attribute	Value
Unit	Case
Rows	644
Primary key	`case_id` (integer)

`wto_cases_v2` joined with `severity_scores_raw` and `case_hs_sections`. Adds `consultation_year`, `is_horizontal_policy`, and severity/HS columns. Cases DS627–DS644 have NaN for severity and HS columns.

7.2 `wto_dyadic_enriched.csv`

Attribute	Value
Unit	Case-directed dyad
Rows	8,373
Primary key	`(case, iso3_1, iso3_2, relationship)`

`wto_dyadic_v2` joined with case attributes. Key added columns:

Variable	Definition
`consultation_year`	Year extracted from `consultations_requested`
`dyadic_severity`	Severity by role: CR→case score; TP-R→0.5; C-TP→0
`is_horizontal_policy`	True for policy cases spanning all 21 HS sections
`disputed_trade_ij_t{0..3}`	Trade from country1→country2 in disputed HS sections at t, t-1, t-2, t-3
`disputed_trade_ji_t{0..3}`	Trade from country2→country1 in disputed HS sections
`disputed_dep_ij_t{0..3}`	Partner sector dependence for country1 exports to country2 in disputed sections

7.3 `ergm_dyad_year_eun.csv`

Attribute	Value
Unit	Directed dyad-year (exporter → importer)
Rows	~545,333
Universe	`bilateral_trade_wto` (all WTO-member pairs, 1995–2024)

Key variable groups:

Group	Variables
Outcome	`has_dispute`, `dispute_count`, `case_id_max`, `case_id_avg`
Severity	`severity_max`, `severity_avg`, `dyadic_severity_max`, `dyadic_severity_avg`
HS sections	`section_1` … `section_21` (dummies from most severe case)
Trade (contemporaneous)	`total_trade_ij`, `export_dependence`, `atopally`, `label`, `depth_index` (from `bilateral_trade_wto`)
Trade lags	`total_trade_ij_t{1..3}`, `export_dependence_t{1..3}`, `atopally_t{1..3}`, `label_t{1..3}`
Disputed sector trade	`disputed_trade_ij_t{0..3}`, `disputed_trade_ji_t{0..3}`
Node attrs (exporter)	44 cols from `country_meta`, suffixed `_1`: `gdppc_1`, `pop_1`, `v2x_polyarchy_1`, `idealpointall_1`, `cum_complainant_1`, `n_complainant_t_1`, etc.
Node attrs (importer)	Same 44 cols suffixed `_2`
TP covariates	`tp_with_j_complainant_i` (has i acted as TP supporting j's complaints), `tp_count_i_supporting_j`

> ***Backward imputation (trade only):*** *1995 → all NaN; 1996 → t-2=t-3=t-1; 1997 → t-3=t-2. Node attributes use contemporaneous year t.*
>
> ***Taiwan:*** *NaN for `idealpointfp_*`, `idealpointall_*`, `idealpointlegacy_*` (not a UN member).*
>
> ***EUN DESTA fix:*** *EUN rows inherit trade agreement data from member states. EUN ATOP = 0 by design (EU is not an ATOP signatory).*

7.4 `ergm_dyad_year_eu_disagg.csv`

EU-disaggregated robustness version. EUN dispute rows replaced by individual EU member state rows:
- **395 EUN-only cases**: expanded to individual EU member state rows present in

`bilateral_trade_wto`
- **37 mixed cases** (EUN + individual members both listed): EUN rows dropped, individual member rows kept
- **30 individual-EU-member-only cases**: unchanged

End of documentation.